

Sreya.V

(908)-392-7667 | Python Developer | New Jersey

sreyavemulapalli9@gmail.com

Profile Summary:

- Experienced Python Developer with a strong background in designing, developing, and deploying scalable applications across industries including banking, healthcare, e-commerce, and telecommunications.
- Proficient in building RESTful APIs, backend services, and web applications using Python frameworks such as Django, Flask, and FastAPI.
- Hands-on experience with cloud platforms (AWS, Azure, GCP) for deploying serverless applications, containerized services, and CI/CD pipelines.
- Strong skills in database design and integration (PostgreSQL, MySQL, MongoDB) with a focus on query optimization and data modeling.
- Expert in writing clean, reusable, and test-driven code with frameworks like PyTest and unittest to ensure maintainability and scalability.
- Experience developing microservices architectures, integrating third-party APIs, and implementing authentication/authorization using OAuth2 and JWT.
- Well-versed in DevOps practices with Git, Docker, Kubernetes, and Jenkins for automated deployments and version control.
- Strong collaborator, engaging with cross-functional teams including product managers, testers, and business stakeholders to deliver high-quality software solutions.

Technical Skills:

Programming Languages	Python (Django, Flask, FastAPI, Pandas, NumPy), JavaScript, SQL, Bash/Shell
Web Frameworks	Django, Flask, FastAPI, Django REST Framework
Databases	PostgreSQL, MySQL, MongoDB, Redis, SQLite
DevOps & Tools	Docker, Kubernetes, Jenkins, Git, GitHub Actions, CI/CD, Terraform
Testing Frameworks	PyTest, UnitTest, Selenium
APIs & Integrations	RESTful APIs, GraphQL, gRPC, OAuth2/JWT authentication
Visualization & BI	Tableau, Power BI, Matplotlib, Seaborn
Version Control	Git, GitHub, GitLab, Bitbucket

PROJECT / WORK EXPERIENCE

Client : UPS , NJ	
Role : Azure Data Engineer	Aug 2024-Present
Title : Enterprise Web Application and API Development	

Led the design and implementation of enterprise-grade web applications and APIs for logistics operations. Focused on low-latency real-time tracking, modular microservices, and analytics integration that improved operational visibility and reduced processing times.

Developed backend services using Django REST Framework and Flask, implementing modular APIs that integrated logistics, tracking, and reporting services across UPS systems.

- Collaborated with front-end teams to design REST APIs, ensuring efficient communication and minimizing response times through caching strategies and optimized queries.
- Integrated third-party APIs (payment gateways, address validation, and SMS notifications) with UPS applications to improve user experience and reduce operational errors.
- Built and deployed containerized microservices with Docker and Kubernetes, scaling applications seamlessly during peak delivery seasons.
- Optimized PostgreSQL queries and implemented indexing strategies, reducing report generation time from minutes to seconds for large datasets.
- Implemented authentication and authorization mechanisms using OAuth2 and JWT, securing sensitive shipment and customer data.
- Wrote test-driven code using pytest and unittest, achieving 95% code coverage and reducing bugs in production releases.
- Deployed CI/CD pipelines using Jenkins and GitHub Actions to automate build, test, and deployment processes across multiple environments.
- Integrated Redis caching layers into Flask applications to handle high-volume requests and improve API response times by 40%.
- Collaborated with data engineering teams to provide Python-based ETL scripts, automating the extraction and transformation of logistics data into analytics dashboards.
- Implemented Celery with RabbitMQ for asynchronous task scheduling, handling millions of background tasks like shipment tracking and notifications daily.
- Migrated legacy Java services into Python-based microservices, simplifying maintenance and improving developer productivity.
- Created automated monitoring scripts in Python to track API uptime, service health, and error logging, reducing downtime incidents by 30%.
- Led daily Agile/Scrum ceremonies, mentoring junior developers and ensuring project milestones were consistently met ahead of deadlines.
- Authored comprehensive technical documentation, onboarding playbooks, and reusable Python libraries for standardized development practices.

- Designed scalable architecture diagrams and deployment workflows for cloud-native applications hosted on AWS and Azure environments.

Client : Bank Of Oklahoma

Role : Python Developer

Sep2023-July2024

Title : Banking Applications & API Development

Responsible for developing secure, compliant Python applications for banking operations, including transaction APIs, serverless workflows, and fraud-detection ETL pipelines. Ensured systems met regulatory standards while optimizing performance and reliability.

- Developed RESTful APIs in Flask and Django to handle secure financial transactions, authentication, and reporting services across multiple banking channels.
- Built serverless applications using AWS Lambda with Python, reducing infrastructure overhead while supporting millions of daily requests.
- Created ETL workflows in Python to process financial transaction logs, applying data validation rules and delivering insights into fraud detection systems.
- Integrated AWS services like S3, RDS, and Redshift for storing and analyzing structured/unstructured datasets in near real-time.
- Implemented role-based authentication in Django applications, ensuring compliance with financial security standards and reducing vulnerabilities.
- Optimized PostgreSQL queries used in banking reports, reducing execution time by 60% during monthly reconciliation processes.
- Designed unit and integration tests with PyTest, ensuring API stability across multiple deployment environments.
- Automated deployment workflows using Jenkins and GitHub Actions for consistent CI/CD pipelines.
- Integrated Tableau dashboards with Python APIs, providing real-time insights for auditors and banking operations teams.
- Developed asynchronous processing scripts using Celery to handle high-volume message queues related to loan processing and approvals.
- Migrated legacy PL/SQL scripts into Python data pipelines, making reporting more efficient and developer-friendly.
- Built monitoring tools in Python to track financial API performance, error rates, and security breaches, sending alerts to operations teams.

- Collaborated with cloud architects to re-engineer financial applications into a microservices-based architecture for scalability.
- Trained junior developers in Python best practices, mentoring them on debugging, testing, and performance optimization.
- Documented detailed API specifications using Swagger/OpenAPI, improving cross-team collaboration and reducing integration errors.
- Worked closely with compliance teams to ensure all Python services adhered to SOX and PCI DSS security regulations.

Client : NextGenPros Inc

Role : Python Developer

March 2022 - Aug 2023

Title : : Healthcare & E-commerce Applications

Built scalable backend services and microservices for healthcare and e-commerce clients, delivering event-driven analytics, robust APIs, and CI/CD-enabled deployments. Worked closely with product teams to prioritize features and improve time-to-market.

- Designed and developed scalable Django applications supporting healthcare and e-commerce clients, implementing secure user management and reporting modules.
- Created real-time event streaming applications in Python integrated with AWS Kinesis and Kafka for analytics dashboards.
- Built microservices with Flask, containerized them using Docker, and deployed to AWS ECS for high-availability workloads.
- Implemented SQLAlchemy ORM models in Python to handle dynamic schema changes and maintain referential integrity in PostgreSQL.
- Integrated third-party APIs for payments, SMS, and email notifications into client-facing Python applications.
- Developed caching strategies with Redis and Memcached, significantly improving the performance of API responses under heavy loads.
- Automated CI/CD deployments using Jenkins pipelines for Django-based healthcare platforms.
- Built custom Python scripts for data cleaning, preprocessing, and reporting, reducing manual tasks by 70%.
- Implemented OAuth2.0 and JWT authentication in APIs, ensuring secure access across all client applications.
- Enhanced user experience by designing RESTful endpoints that allowed seamless integration with React frontends.
- Used Pandas and NumPy for processing healthcare datasets, delivering clean structured data to analytics teams.

- Deployed scalable serverless APIs using AWS API Gateway and Lambda, reducing costs while maintaining flexibility.
- Collaborated with DevOps engineers to build Kubernetes clusters for microservices hosting, ensuring zero downtime deployments.
- Led code reviews for Python development teams, ensuring adherence to PEP8 standards and best practices.
- Developed logging and monitoring services using Python, integrated with CloudWatch and ELK stack.
- Provided mentorship to junior developers, organizing weekly coding workshops on Python and Django.

Client : FTX , India

Role :Senior Python Developer

Jan2020-Feb2022

Title : Financial Trading & Analytics Platforms

Developed high-performance trading and analytics systems tailored for financial markets, emphasizing low-latency data ingestion, resilient order processing, and real-time dashboards. Implemented strict security, monitoring, and compliance controls.

- Built high-performance trading and analytics applications using Python, Django, and Flask for financial clients.
- Developed real-time market data ingestion services, integrating APIs from exchanges and handling millions of records daily.
- Implemented Redis for caching real-time financial data, improving response times for trading dashboards.
- Created custom Python modules for financial calculations, portfolio management, and risk analysis.
- Deployed containerized services in Kubernetes clusters with monitoring and alerting systems for trading applications.
- Designed automated trading scripts in Python, integrated with REST and WebSocket APIs for low-latency transactions.
- Implemented secure login and multi-factor authentication features for web-based financial dashboards.
- Developed Celery task queues to handle high-frequency background jobs like order matching and settlement processing.
- Optimized MongoDB queries for transaction logs, enabling faster reconciliation processes for compliance teams.
- Built automated test frameworks using PyTest, ensuring high-quality production releases with minimal errors.

- Developed APIs in Flask to expose financial data and reports to external clients securely.
- Integrated with Kafka streams to manage high-frequency trading data pipelines.
- Provided technical guidance to junior developers on building scalable and secure Python services.
- Participated in Agile ceremonies and actively collaborated with product managers to prioritize features.
- Documented trading platform architecture, creating detailed diagrams and system specifications.
- Ensured all Python applications complied with security and encryption standards required by financial regulations.

Client : GormalOne, Bangalore

Role : Python Developer

Jan 2019 - Dec2019

Title : Customer Analytics & Reporting Applications

Developed customer analytics platforms and reporting applications, handling end-to-end data ingestion, API development, and dashboarding. Modernized legacy systems by migrating to Django-based solutions and improving data pipeline reliability.

- Developed customer analytics applications using Django, enabling real-time reporting on customer engagement trends.
- Designed data ingestion scripts in Python to collect data from APIs, CSVs, and relational databases.
- Implemented visualization dashboards using Django and Chart.js, providing management with actionable insights.
- Created REST APIs to expose analytics results to business partners in retail and e-commerce sectors.
- Used Pandas and NumPy for preprocessing structured and semi-structured customer datasets.
- Built Flask microservices to support marketing automation features such as targeted notifications.
- Implemented asynchronous job scheduling with Celery and RabbitMQ, handling campaign workflows efficiently.
- Integrated with NoSQL databases like MongoDB and Redis for customer profile storage and caching.
- Deployed Python applications to AWS EC2 instances and automated updates with Ansible.
- Enhanced API performance by using Redis caching and optimizing query execution.
- Collaborated with business teams to design analytics reports, aligning technical deliverables with KPIs.
- Built logging systems in Python, integrated with ELK stack for system observability.
- Migrated legacy PHP applications into Django, simplifying maintenance and improving security.

- Created reusable Django templates and libraries to standardize reporting modules across multiple projects.
- Wrote unit and functional tests for APIs using PyTest, ensuring application stability.
- Documented APIs, workflows, and database schemas for easier onboarding of new developers.

Education:

- Master's in Computer Science from New Jersey Institute of Technology.
- Bachelor's in Computer Science from SRM University.